



Silent SOS – AI-Powered Intelligent Emergency Response System

Every Second Matters. Every Life Counts.

Team: Aurora Response • Tech: Flutter, Node.js, AWS, Gemini AI • Style: Dark UI • Visual: Futuristic maps & live dashboards

TEAM MEMBERS DETAILS :

D.V.L.RAGHAVA ; D.NIHARIKA TEJASWI;CH.SOWMYA;K.VYHSNAVI;B.CHANTI

The Problem — Real-world Emergency Gaps



Delayed response times

Average dispatch delays, traffic and manual triage cost critical minutes.



Victims unable to call

Unconscious or immobile victims cannot trigger help manually.



Women & elderly safety

High-risk demographics face disproportionate delays and danger.



Network & manual-reporting limits

Spotty connectivity and slow manual reports block timely response.

The result: missed minutes → increased mortality and long-term harm.



Solution – Silent SOS

Autonomous emergency detection and response: sensor intelligence + on-device AI + cloud orchestration. Detects falls, abnormal vitals, and suspicious audio; auto-notifies contacts and dispatchers with precise location and incident summary.

01

On-device sensing

Accelerometer, gyroscope, microphone, battery & network telemetry.

02

AI risk analysis

Local model scores risk; compresses incident payload.

03

Cloud orchestration

Gemini + backend validate, store, and broadcast alerts.

04

Live response

WebSocket updates to dashboard, emergency contacts, and first responders.

Key Features



Voice Activated SOS

Hands-free activation via wake phrase or distress keywords.



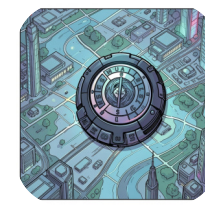
AI Risk Detection

Multi-sensor fusion for contextual assessment and prioritisation.



Automatic Fall Detection

Instant detection with false-positive suppression.



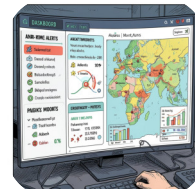
Live GPS Tracking

High-frequency location updates with accuracy estimate.



Background Monitoring

Low-power background services preserve battery and privacy.



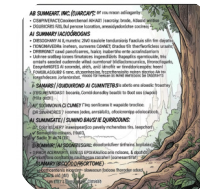
Real-Time Dashboard

Control centre for responders: map, replay, device status, AI reports.



Emergency Contacts

Multi-channel notifications: SMS, call, push, and third-party APIs.



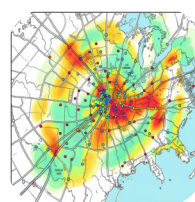
AI Incident Summary

Concise incident reports with priority, probable cause, and recommendations.



Cloud Storage & Video Upload

Encrypted media uploads for evidence and post-incident analysis.

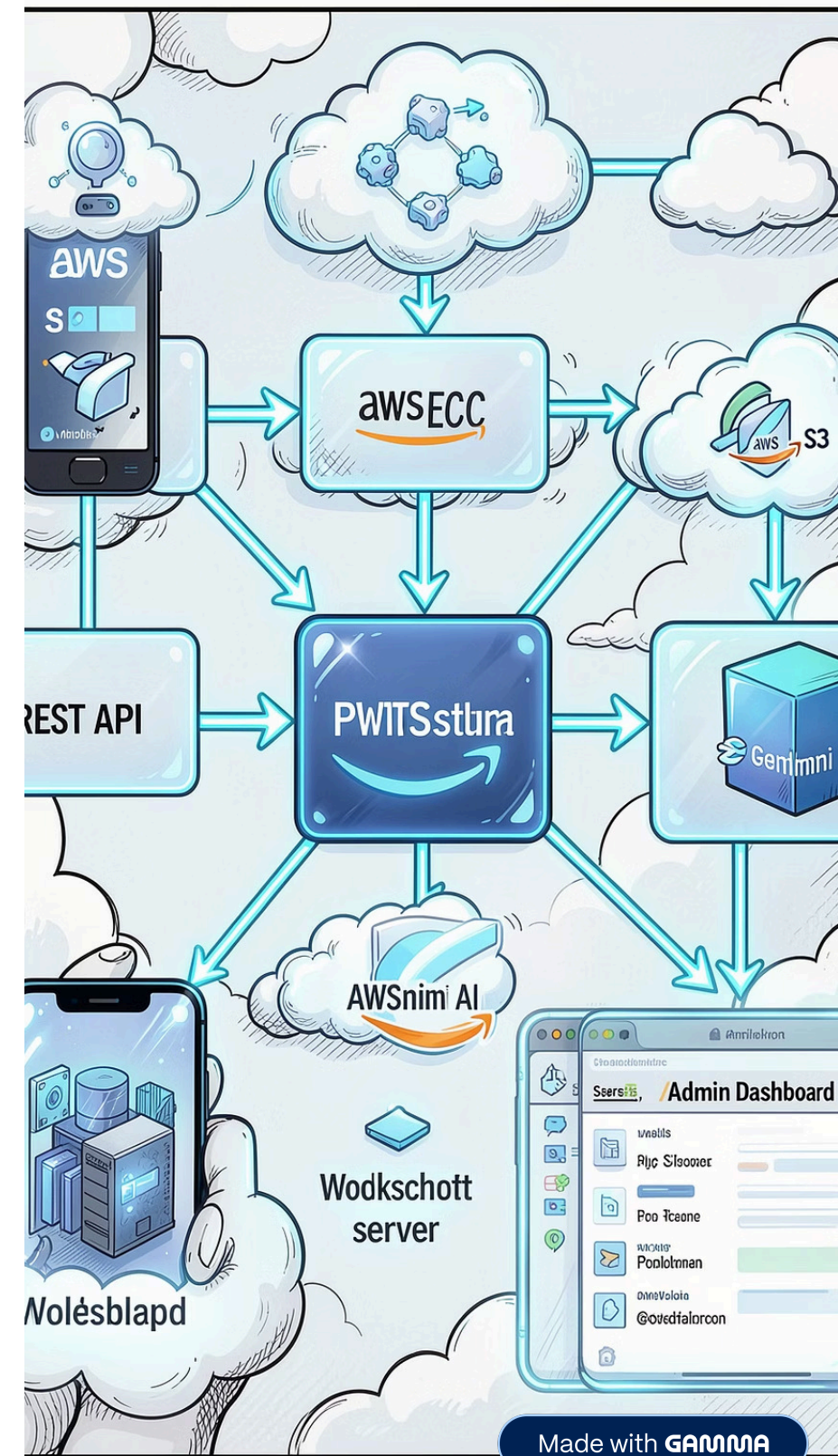
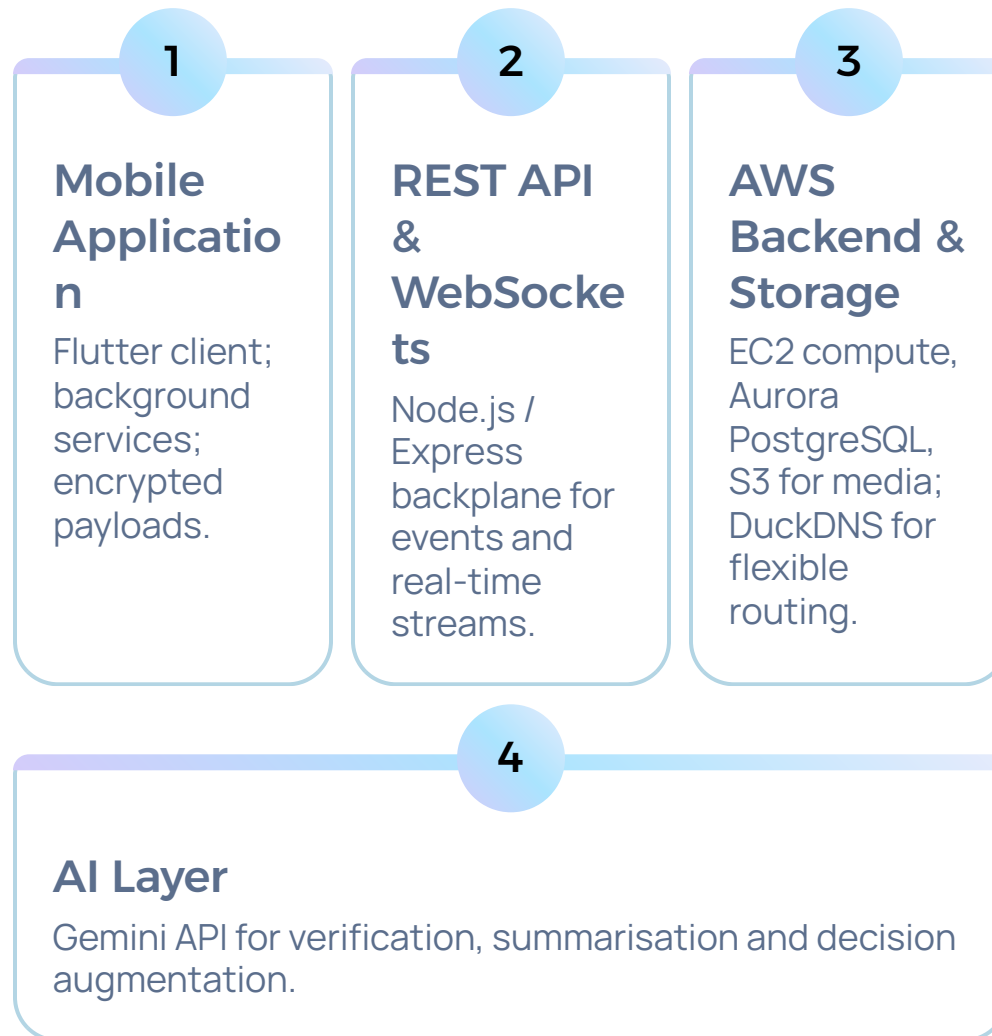


Live Map & Notifications

Real-time geospatial alerts and severity heatmaps for operators.

Fully cloud-native flow with low-latency WebSockets and secure storage. Designed for horizontal scaling and fault tolerance.

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Technology Stack



Frontend

Flutter • Provider • Dart
— cross-platform with
native performance.



Database

Amazon Aurora
PostgreSQL — high
availability & read
replicas.



AI

Google Gemini API —
incident verification,
summary &
recommendations.



Version Control

Git • GitHub — CI/CD
pipelines for production
releases.



Backend

Node.js • Express • REST
APIs • WebSocket
realtime layer.



Cloud

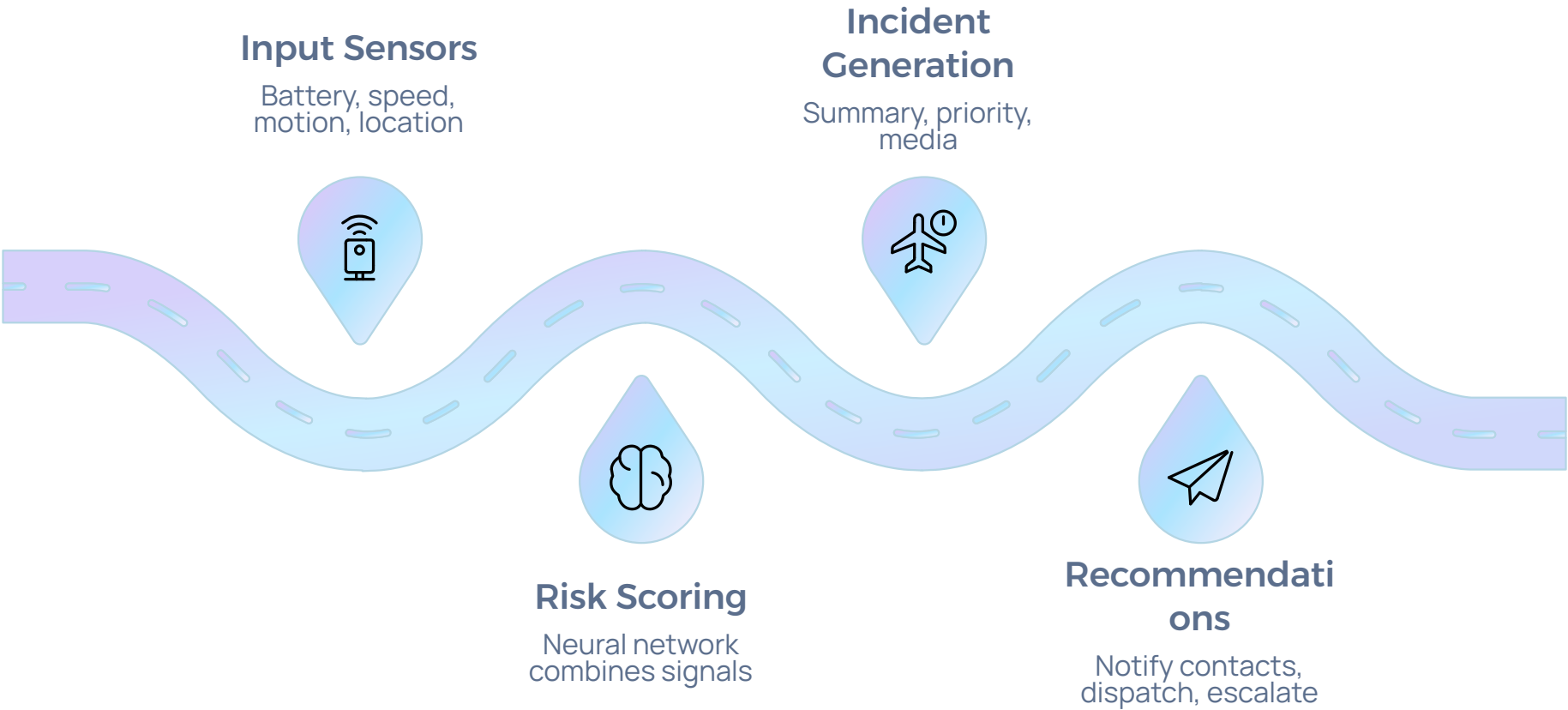
AWS EC2, S3, DuckDNS —
secure, scalable, cost-
optimised.



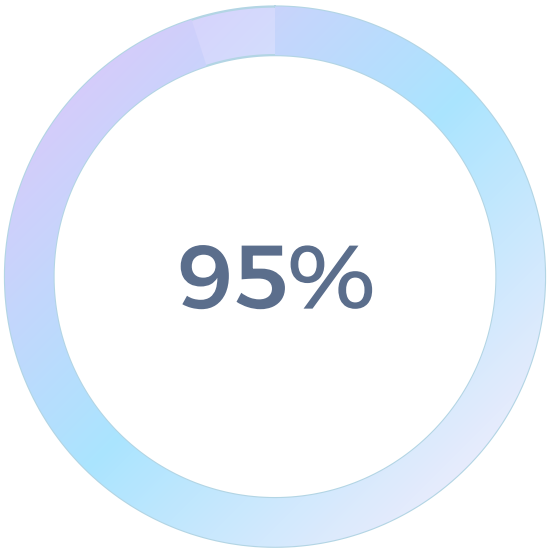
Maps & Dashboard

React • Leaflet •
OpenStreetMap •
WebSocket-driven UI.

AI Intelligence Module



On-device preprocessing reduces bandwidth; Gemini verifies and enriches incident summaries. The model outputs a continuous risk score and discrete incident labels for rapid routing.



Detection reliability (simulated)



Average on-device inference time



Median alert transmission latency

Key signals: motion patterns, fall signatures, audio anomalies, battery & network context. Privacy preserved via local inference and selective upload.

Live Dashboard & Control Centre

Live Map

Device clusters, heatmaps, geofenced zones, and replay capability.

Connected Devices

Real-time status, battery, network, and sensor telemetry.

Alerts & Replay

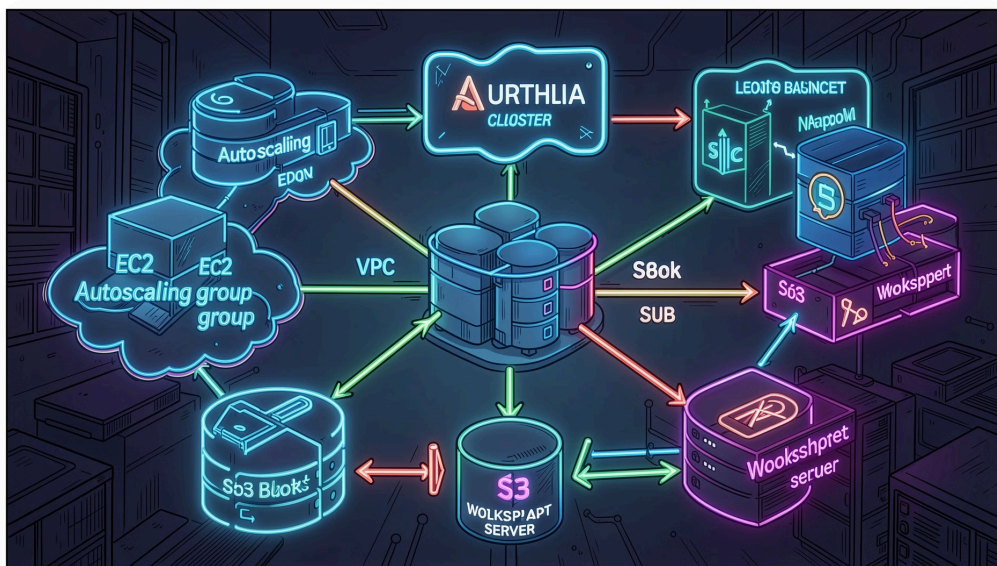
Chronological event stream with media and AI summaries.

AI Reports

Priority, probable cause, recommended actions and responder assignment.



Production Backend & Deployment



Production considerations:

- Autoscaling EC2 pools behind LB, health-checked WebSocket servers.
- Aurora for ACID storage, read replicas for dashboard queries.
- S3 for encrypted media retention and evidence lifecycle.
- Secure APIs with JWT, rate limiting, and audit logs.
- CI/CD pipelines, blue/green deployments, and post-deploy smoke tests.

☐ Design for sub-second alert propagation and 99.95% availability for critical operations.

Roadmap – Impact & Future Scope

Q3 – Wearables & Watch Support

Low-power SDK for smartwatches and BLE wearables.

1

2

Q4 – Drone & Hospital Integration

Automated drone dispatch trials and EHR hooks for hospitals.

3

Next Year – Mesh & Satellite

Offline mesh networking and satellite fallback for connectivity blackspots.

4

Ongoing – Model Improvements

Continual ML updates to reduce false positives and personalise thresholds.

Conclusion

Silent SOS is an AI-first, cloud-native life-saving platform built to reduce response times, automate incident detection, and empower responders with precise, actionable intelligence.

Saving Lives Through Intelligence.

Thank you – we welcome questions and pilot partnerships.